
PE-EC801B: Fiber Optic Communication

Quick-Recall One-Liners Revision Guide

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Chapter 1: Introduction

1. **Light carrier frequency range** → Terahertz range (10^{14} Hz).
2. **Physical principle of light confinement in fiber core** → Total Internal Reflection (TIR).
3. **Snell's Law equation** → $n_1 \sin \theta_1 = n_2 \sin \theta_2$.
4. **Critical angle formula at core-clad interface** → $\theta_c = \sin^{-1}(n_2/n_1)$.
5. **Two necessary conditions for TIR** → Denser to rarer medium propagation; incidence angle exceeding critical angle.
6. **Acceptance angle definition** → Maximum incident angle at air-core interface for guided propagation.
7. **Numerical Aperture (NA) definition** → Sine of acceptance angle in air, representing light-gathering capacity.
8. **NA formula in terms of refractive indices** → $NA = \sqrt{n_1^2 - n_2^2}$.
9. **Approximate NA formula in terms of index difference (Δ)** → $NA \approx n_1 \sqrt{2\Delta}$.
10. **Relative refractive index difference (Δ) formula** → $\Delta = \frac{n_1 - n_2}{n_1}$.
11. **Rays passing through fiber's longitudinal central axis** → Meridional rays.
12. **Rays following helical paths without crossing central axis** → Skew rays.
13. **Acceptance angle formula for air launch ($n_0 = 1$)** → $\theta_a = \sin^{-1}(\sqrt{n_1^2 - n_2^2})$.
14. **Geometric (Ray) model validity condition** → Core diameter much larger than wavelength ($d \gg \lambda$).
15. **Wave model mandatory condition** → Core diameter comparable to wavelength ($d \approx \lambda$).
16. **Non-propagating mode in cylindrical dielectric rods** → Transverse Electromagnetic (TEM) mode.
17. **Mode class where $E_z = 0$ and $H_z \neq 0$** → Transverse Electric (TE) mode.
18. **Mode class where $H_z = 0$ and $E_z \neq 0$** → Transverse Magnetic (TM) mode.
19. **Mode class where both $E_z \neq 0$ and $H_z \neq 0$** → Hybrid mode (HE or EH).
20. **Cause of hybrid modes in fibers** → Vector boundary conditions at curved core-cladding interface.
21. **Main advantages of fiber over copper** → Enormous bandwidth, complete immunity to EMI, and low attenuation.
22. **Key drawbacks of optical fiber cabling** → Fragility, high initial installation cost, and splicing difficulty.
23. **Typical cladding diameter size in standard fibers** → $125 \mu\text{m}$.
24. **Launching medium refractive index (air)** → $n_0 = 1.0$.
25. **Critical angle for $n_1 = 1.82$ and $n_2 = 1.73$** → 71.91° .

Chapter 2: Optical Fibers

1. **Step-index refractive index profile definition** → Constant core index dropping abruptly at boundary.
2. **Graded-index refractive index profile formula** → $n(r) = n_1 \sqrt{1 - 2\Delta(r/a)^\alpha}$ for $r < a$.
3. **Profile parameter value (α) for parabolic graded-index** → $\alpha = 2$.
4. **Light path trajectory in step-index fiber** → Zig-zag paths via discrete reflections (TIR).
5. **Light path trajectory in graded-index fiber** → Continuous sinusoidal curving refraction paths.
6. **Core diameter of standard single-mode fiber (SMF)** → 8 to 10 μm .
7. **Core diameter of standard multimode fiber (MMF)** → 50 to 100 μm .
8. **Intermodal dispersion in single-mode fiber** → Zero.
9. **Normalized frequency (V -number) formula** → $V = \frac{2\pi a}{\lambda} \text{NA}$.
10. **Single-mode propagation condition for step-index fiber** → $V < 2.405$.
11. **First guided mode to propagate in optical fiber** → Fundamental HE_{11} mode.
12. **Number of guided modes (N) in step-index multimode fiber** → $N \approx \frac{V^2}{2}$.
13. **Number of guided modes (N) in graded-index parabolic fiber** → $N \approx \frac{V^2}{4}$.
14. **Cutoff wavelength (λ_c) formula for single-mode fiber** → $\lambda_c = \frac{2\pi a \text{NA}}{2.405}$.
15. **Mode-Field Diameter (MFD) definition** → Radial width where optical power density falls to $1/e^2$.
16. **MARCUSE formula purpose** → Estimating mode-field diameter in step-index fibers.
17. **Field distribution function in the core ($r < a$)** → Bessel function of the first kind (J_ν).
18. **Field distribution function in the cladding ($r > a$)** → Modified Bessel function of the second kind (K_ν).
19. **Boundary continuity field components at $r = a$** → Tangential components E_z , H_z , E_ϕ , H_ϕ .
20. **Number of guided modes for $V = 75.78$ in step-index** → 2871.
21. **Cutoff wavelength for $a = 4 \mu\text{m}$, $n_1 = 1.48$, $\Delta = 0.2\%$** → 978 nm.
22. **Key source compatible with single-mode fibers** → Laser Diode (LD).
23. **Key disadvantage of step-index multimode fiber** → High intermodal dispersion limiting bandwidth.
24. **Key advantage of graded-index fiber over step-index MMF** → Lower intermodal dispersion.
25. **MFD significance** → Determines coupling efficiency, splice sensitivity, and microbending losses.

Chapter 3: Signal Degradation

1. **Attenuation definition** → Reduction in optical signal power during propagation.
2. **Attenuation coefficient (α) formula** → $\alpha = \frac{10}{L} \log_{10}(P_{in}/P_{out}) \dots \text{dB/km}$.
3. **Signal dispersion definition** → Temporal broadening of light pulses during propagation.
4. **Consequence of dispersion on digital transmission** → Inter-Symbol Interference (ISI)

limiting bandwidth.

5. **Dominant scattering mechanism in optical fibers** → Rayleigh scattering.
6. **Rayleigh scattering coefficient wavelength dependence** → $\alpha_{sc} \propto 1/\lambda^4$.
7. **Rayleigh scattering physical cause** → Microscopic density and composition fluctuations frozen in glass.
8. **Intrinsic absorption edges in silica** → UV electronic absorption; IR molecular vibrational absorption.
9. **Extrinsic absorption causes in fibers** → Transition metal impurities and hydroxyl (OH^-) ions.
10. **Peak absorption wavelengths of water (OH^-) ions** → 950 nm, 1240 nm, and 1380 nm.
11. **Attenuation value in first transmission window (850 nm)** → 2.0 to 3.0 dB/km.
12. **Attenuation value in second transmission window (1310 nm)** → 0.4 dB/km.
13. **Attenuation value in third transmission window (1550 nm)** → 0.2 dB/km.
14. **Unique dispersion characteristic of standard SMF at 1310 nm** → Zero chromatic dispersion.
15. **Key advantage of 1550 nm transmission window** → Minimum attenuation window matching EDFA band.
16. **Macrobending loss cause** → Fiber curvature radius of a few centimeters.
17. **Microbending loss cause** → Microscopic axial deviations from mechanical lateral stress.
18. **Cabling primary functions** → Mechanical protection, environmental shielding, and microbending mitigation.
19. **Intermodal delay difference (Δt_{modal}) formula for step-index** → $\Delta t_{\text{modal}} \approx \frac{n_1 L}{c} \Delta$.
20. **RMS pulse broadening (σ_{modal}) formula for step-index MMF** → $\sigma_{\text{modal}} \approx \frac{n_1 L \Delta}{2\sqrt{3}c}$.
21. **Chromatic dispersion components** → Material dispersion and waveguide dispersion.
22. **Material dispersion formula coefficient (D_M)** → $D_M = -\frac{\lambda}{c} \frac{d^2 n_1}{d\lambda^2}$.
23. **Dispersion-Shifted Fiber (DSF) design objective** → Shift zero-dispersion wavelength to 1550 nm.
24. **Dispersion-Flattened Fiber (DFF) design objective** → Keep dispersion low ($< 2 \text{ ps}/(\text{nm} \cdot \text{km})$) across 1300–1600 nm.
25. **Preform fabrication dry mixture reactants** → SiCl_4 , GeCl_4 , and O_2 .
26. **Sintering definition in MCVD** → Heating deposited soot to form solid glass layers.
27. **Preform fabrication method using concentric inner/outer pots** → Double-Crucible method.
28. **OTDR full form** → Optical Time Domain Reflectometer.
29. **OTDR continuous downward slope represents** → Rayleigh backscattering attenuation.
30. **OTDR sharp upward peaks represent** → Fresnel reflections from connectors or faults.
31. **Standard method for high-accuracy fiber attenuation measurement** → Cut-Back method.
32. **GRIN-rod lens diameter range** → 0.5 to 2 mm.
33. **GRIN-rod lens divergence angle** → 1 to 5 degrees.

34. **GRIN-rod lens fiber alignment position** → Exactly at the focal length of the lens.
35. **Typical insertion loss of expanded-beam connector** → 0.7 dB.

Chapter 4: Optical Sources & Detectors

1. **LED light emission process** → Spontaneous emission of photons.
2. **Laser diode light emission process** → Stimulated emission of coherent photons.
3. **LED spectral width ($\Delta\lambda$) range** → Broad (30 to 50 nm).
4. **Laser diode spectral width ($\Delta\lambda$) range** → Narrow (< 1 to 3 nm).
5. **Condition where excited state population exceeds ground state** → Population inversion.
6. **Method of achieving population inversion at junction** → Heavy doping of both regions under forward bias.
7. **Lasing threshold gain (g_{th}) formula** → $g_{th} = \alpha_t + \frac{1}{2L} \ln \left(\frac{1}{R_1 R_2} \right)$.
8. **Reflectivity (R) formula at normal incidence** → $R = \left(\frac{n-1}{n+1} \right)^2$.
9. **Reflectivity of GaAs-air interface ($n = 3.6$)** → 32% (or 0.32).
10. **Longitudinal mode count (m) formula** → $m = \frac{2nL}{\lambda}$.
11. **Mode count for ruby laser ($L = 3$ cm, $n = 1.6$, $\lambda = 0.43$ μm)** → 2.23×10^5 .
12. **Frequency separation of modes (Δf) formula** → $\Delta f = \frac{c}{2nL}$.
13. **Mode frequency separation for $L = 5$ cm, $n = 1.8$** → 1.67 GHz.
14. **Wavelength mode separation ($\Delta\lambda$) formula** → $\Delta\lambda = \frac{\lambda^2 \Delta f}{c}$.
15. **Mode separation for $\lambda = 0.5$ μm and $\Delta f = 2$ GHz** → 1.67×10^{-12} m.
16. **Radiative minority carrier lifetime (τ_r) formula** → $\tau_r = \frac{1}{B \cdot p_0}$.
17. **Radiative lifetime for $p_0 = 10^{18}$ cm^{-3} , $B = 7.21 \times 10^{-10}$ $\text{cm}^3 \text{s}^{-1}$** → 1.39 ns.
18. **Shorter wavelength semiconductor material system** → GaAs/AlGaAs DH.
19. **Recombination characteristics of indirect band-gap semiconductors** → Slow and heat-producing.
20. **PIN photodiode internal gain (M)** → $M = 1$ (no internal gain).
21. **Avalanche Photodiode (APD) gain mechanism** → Impact ionization under high reverse bias.
22. **APD typical reverse bias operating voltage range** → 100 to 300 V.
23. **Quantum efficiency (η) definition** → Fraction of incident photons generating collected carrier pairs.
24. **Responsivity (R) definition** → Ratio of output photocurrent to incident optical power.
25. **Responsivity formula in terms of quantum efficiency and wavelength** → $R = \frac{\eta e \lambda}{h c} \approx \frac{\eta \lambda (\mu\text{m})}{1.24}$ A/W.
26. **Reason Silicon is unsuitable for 1550 nm window** → Bandgap too wide ($E_g = 1.1$ eV), transparent to 1550 nm.
27. **Preferred photodetector material for 1310/1550 nm windows** → Indium Gallium Arsenide (InGaAs).
28. **Primary noise sources in digital optical receiver** → Shot noise, thermal noise, and

amplifier noise.

29. **Thermal noise (Johnson noise) mean-square current formula** $\rightarrow \langle i_{thermal}^2 \rangle = \frac{4k_B T B}{R_L}$.
30. **APD shot noise mean-square current formula** $\rightarrow \langle i_{shot}^2 \rangle = 2e(I_p + I_d)M^2 F(M)B$.
31. **Link Power Budget equation** $\rightarrow P_T - P_R = \alpha L + N_{sp}\alpha_{sp} + N_c\alpha_c + M_s$.
32. **System rise-time (t_{sys}) budget equation** $\rightarrow t_{sys} = (t_{tx}^2 + t_{mat}^2 + t_{mod}^2 + t_{rx}^2)^{1/2}$.
33. **Maximum rise-time limit for NRZ digital systems** $\rightarrow t_{sys} \leq 0.7T$.
34. **Quantum limit of detection for BER = 10^{-9}** $\rightarrow$ Average minimum of 21 photons per pulse.
35. **Recombination lifetime of direct band-gap semiconductors** $\rightarrow$ Nanoseconds.

Chapter 5: Optical Switches & Amplifiers

1. **EDFA full form** $\rightarrow$ Erbium-Doped Fiber Amplifier.
2. **EDFA gain medium** $\rightarrow$ Silica fiber doped with Erbium (Er^{3+}) ions.
3. **EDFA standard pumping wavelengths** $\rightarrow$ 980 nm and 1480 nm.
4. **Raman amplifier gain mechanism** $\rightarrow$ Stimulated Raman Scattering (SRS).
5. **Raman amplifier gain medium** $\rightarrow$ Standard transmission fiber core (distributed amplification).
6. **Raman pump wavelength condition** $\rightarrow$ Must be ≈ 100 nm shorter than signal.
7. **Stokes shift frequency value in silica** $\rightarrow \approx 13.2$ THz.
8. **WDM full form** $\rightarrow$ Wavelength Division Multiplexing.
9. **Channel spacing in Coarse WDM (CWDM)** $\rightarrow$ Wide, typically 20 nm.
10. **Channel spacing in Dense WDM (DWDM)** $\rightarrow$ Narrow, ≤ 1.6 nm (typically 0.8 nm/100 GHz).
11. **Fused Biconical Taper (FBT) power splitting mechanism** $\rightarrow$ Evanescent wave coupling in tapered region.
12. **Symmetric directional coupler power transfer equation ($P_2(z)$)** $\rightarrow P_2(z) = P_0 \sin^2(\kappa z)$.
13. **Coupling length (L_c) formula for symmetric coupler** $\rightarrow L_c = \frac{\pi}{2\kappa}$.
14. **Switch state when directional coupler is phase-matched ($\Delta\beta = 0$)** $\rightarrow$ Cross state (100% power transfer).
15. **Switch state when phase-mismatch ($\Delta\beta \neq 0$) is induced by voltage** $\rightarrow$ Bar state.
16. **Electro-optic switch material substrate** $\rightarrow$ Lithium Niobate (LiNbO_3).
17. **Mach-Zehnder Interferometer (MZI) switch switching voltage symbol** $\rightarrow V_\pi$.
18. **Four protocol layers of SONET** $\rightarrow$ Photonic, Section, Line, and Path layers.
19. **OADM function in optical networks** $\rightarrow$ Selectively drop and add specific wavelengths at nodes.
20. **OXC function in optical networks** $\rightarrow$ Route wavelengths dynamically between multiple fiber ports.
21. **RWA problem full form** $\rightarrow$ Routing and Wavelength Assignment.
22. **Constraint requiring a lightpath to use same wavelength throughout** $\rightarrow$ Wavelength continuity constraint.
23. **Device that removes wavelength continuity constraint** $\rightarrow$ Wavelength converter.

24. **Typical noise figure of EDFA** → 3 to 5 dB.
25. **Number of fibers required for unidirectional WDM duplex link** → Two.

Chapter 6: Nonlinear Effects

1. **Optical Kerr effect definition** → Refractive index changes linearly with local optical intensity.
2. **Refractive index formula under Kerr effect** → $n(I) = n_0 + n_2 I$.
3. **Nonlinear refractive index coefficient (n_2) of silica** → $\approx 2.6 \times 10^{-20} \text{ m}^2/\text{W}$.
4. **Kerr-effect-based nonlinear phenomena** → Self-Phase Modulation, Cross-Phase Modulation, Four-Wave Mixing.
5. **Inelastic scattering-based nonlinear phenomena** → Stimulated Raman Scattering and Stimulated Brillouin Scattering.
6. **Physical mechanism of SBS** → Light scattering from acoustic density waves (acoustic phonons).
7. **Physical mechanism of SRS** → Light scattering from optical molecular vibrations (optical phonons).
8. **Direction of scattered wave propagation in SBS** → Strictly backward (backscattering).
9. **Direction of scattered wave propagation in SRS** → Both forward and backward directions.
10. **Frequency shift value in SBS** → Small, ≈ 10 to 11 GHz.
11. **Frequency shift value in SRS** → Large, ≈ 13.2 THz.
12. **Typical SBS threshold power level in standard SMF** → Very low (1 to 10 mW).
13. **Typical SRS threshold power level in standard SMF** → Very high (≈ 1 W).
14. **Effective fiber interaction length (L_{eff}) formula** → $L_{eff} = \frac{1 - e^{-\alpha L}}{\alpha}$.
15. **SBS threshold power formula** → $P_{th,SBS} \approx \frac{21 A_{eff}}{g_B L_{eff}}$.
16. **SRS threshold power formula** → $P_{th,SRS} \approx \frac{16 A_{eff}}{g_R L_{eff}}$.
17. **Key physical consequence of Self-Phase Modulation (SPM)** → Spectral broadening of the pulse.
18. **SPM nonlinear phase shift (ϕ_{NL}) formula** → $\phi_{NL}(t) = \gamma P(t)z$.
19. **Fiber nonlinearity parameter (γ) formula** → $\gamma = \frac{2\pi n_2}{\lambda A_{eff}}$.
20. **Cross-Phase Modulation (XPM) definition** → Pulse phase modulation induced by co-propagating channel intensities.
21. **Four-Wave Mixing (FWM) definition** → Third-order parametric process generating sidebands from channel beating.
22. **Key condition for efficient FWM generation** → Phase-matching condition ($\Delta k \approx 0$).
23. **Fiber design type that mitigates FWM in DWDM** → Non-Zero Dispersion-Shifted Fiber (NZ-DSF).
24. **Optical soliton definition** → Pulse that propagates without broadening using GVD and SPM.
25. **Balancing mechanism of fundamental solitons** → Anomalous dispersion ($D > 0$) GVD cancels SPM red-shift chirp.

- 26. **Pulse shape of fundamental soliton** → Hyperbolic secant (sech).
- 27. **Equation governing soliton propagation in fibers** → Nonlinear Schrödinger Equation (NLSE).
- 28. **GVD parameter (β_2) relation to dispersion parameter (D)** → $D = -\frac{2\pi c}{\lambda^2} \beta_2$.
- 29. **Regime where fundamental solitons can propagate** → Anomalous dispersion regime ($\beta_2 < 0, D > 0$).
- 30. **Jitter caused by soliton-amplifier noise interaction** → Gordon-Haus jitter.